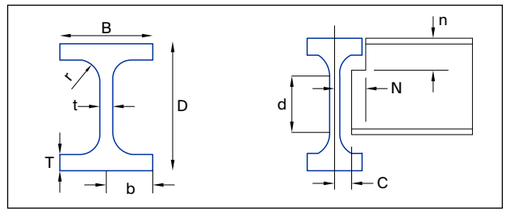
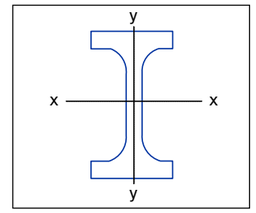
2.3
Universal Beams & Columns (UB & UC)



Dimensions	EN10365 / ASTM A6
Specification	EN10025 / ASTM A36 / ASTM A572
Size Range	UB: 5”x3” (127mm x 76mm) to 44”x16” (1118mm x 406mm) UC: 4”x4” (102mm x 102mm) to 14”x16” (356mm x 406mm)



Imperial Units

Designation			Depth of Section	Width of Section	Thickness		Root Radius	Depth between Fillets	Area of Section	Ratios for Local Buckling		Dimensions for Detailing		
Size	Mass per Metre				Flange	Web				Flange	Web	End clearance	Notch	
			D	B	T	t	r	d	A	b/T	d/t	C	N	n
in (mm)	lb/ft	kg/m	mm	mm	mm	mm	mm	mm	cm²			mm	mm	mm
W14-14x16 (356x406)	176	261.9	386.6	397.5	33.3	21.1	15.2	289.7	334	5.97	13.7	13	198	48
	193	287.1	393.6	399.0	36.5	22.6	15.2	290.2	366	5.47	12.8	13	198	52
	211	314	399.3	401.3	39.6	24.9	15.2	289.7	400	5.06	11.6	14	198	55
	219	326	403.1	402.0	41.2	25.5	15.2	290.3	415	4.88	11.4	15	198	56
	228	339.9	406.4	403.0	42.9	26.6	15.2	290.2	433	4.70	10.9	15	198	58
	233	346.7	407.4	403.6	43.7	27.2	15.2	289.6	442	4.62	10.7	16	198	59
	257	382.5	416.1	406.3	48.0	29.8	15.2	289.7	488	4.23	9.71	17	198	63
	264	393	419.0	407.0	49.2	30.6	15.2	290.2	501	4.14	9.48	17	198	64
	283	421.1	425.2	409.2	52.6	32.8	15.2	289.6	537	3.89	8.84	18	198	68
	311	462.8	434.8	412.2	57.4	35.8	15.2	289.6	590	3.59	8.09	20	198	73
	314	467	436.6	412.2	58.0	35.8	15.2	290.2	595	3.55	8.11	20	198	73
	342	509	445.5	415.5	62.7	39.1	15.2	289.6	649	3.31	7.40	22	198	78
	370	551	455.6	418.5	67.5	42.1	15.2	290.2	702	3.10	6.89	23	198	83
	398	592.3	464.6	421.4	72.3	45.0	15.2	289.7	755	2.92	6.44	24	198	87
	426	633.9	474.6	424.0	77.0	47.6	15.2	290.2	808	2.75	6.10	26	198	92
	455	677.1	483.1	427.6	81.5	51.2	15.2	289.6	863	2.62	5.66	28	198	97
	500	744.1	497.8	432.1	88.9	55.6	15.2	289.6	948	2.43	5.21	30	198	104
	550	818.5	514.1	436.9	97.0	60.5	15.2	289.6	1043	2.25	4.79	32	198	112
	605	900.3	531.4	442.3	105.6	65.9	15.2	289.8	1147	2.09	4.40	35	198	121
	665	989.6	549.7	448.3	114.8	71.9	15.2	289.7	1261	1.95	4.03	38	198	130
	730	1086	569.5	454.4	124.7	78.0	15.2	289.7	1385	1.82	3.72	41	198	140
W16-16x5 1/2 (406x140)	26	39.0	398.0	141.8	8.6	6.4	10.2	360.4	49.7	8.24	56.3	5	78	19
	31	46.0	403.2	142.2	11.2	6.8	10.2	360.4	58.6	6.35	53.0	5	78	21
W16-16x7 (406x178)	36	54.1	402.6	177.7	10.9	7.7	10.2	360.4	69.0	8.15	46.8	6	95	21
	40	60.1	406.4	177.9	12.8	7.9	10.2	360.4	76.5	6.95	45.6	6	95	23
	45	67.1	409.4	178.8	14.3	8.8	10.2	360.4	85.5	6.25	41.0	6	95	25
	50	74.2	412.8	179.5	16.0	9.5	10.2	360.4	94.5	5.61	37.9	7	95	26
	57	84.8	417.3	180.8	18.2	10.9	10.2	360.6	108	4.98	33.0	7	95	28
W16-16x10 1/4 (406x260)	67	99.7	414.8	260.0	16.9	10.0	10.2	360.6	127	7.70	36.0	7	135	27
	77	114.6	419.6	261.5	19.3	11.6	10.2	360.6	146	6.77	31.2	8	135	30
	89	132.4	425.4	263.3	22.2	13.3	10.2	360.6	169	5.92	27.0	9	135	32
	100	148.8	431.0	264.8	25.0	14.9	10.2	360.6	190	5.29	24.3	9	135	35
W18-18x6 (457x152)	35	52.3	449.8	152.4	10.9	7.6	10.2	407.6	66.6	6.99	53.6	6	82	21
	40	59.8	454.6	152.9	13.3	8.1	10.2	407.6	76.2	5.75	50.3	6	82	24
	45	67.2	458.0	153.8	15.0	9.0	10.2	407.6	85.6	5.13	45.3	7	82	25

Surface Area	Second Moment of Area		Radius of Gyration		Elastic Modulus		Plastic Modulus		Buckling Parameter	Torsional Index	Warping Constant	Torsional Constant
Per metre	Axis x-x	Axis y-y	Axis x-x	Axis y-y	Axis x-x	Axis y-y	Axis x-x	Axis y-y				
m²	cm⁴	cm⁴	cm	cm	cm³	cm³	cm³	cm³	u	x	H	J
2.29	89040	34860	16.3	10.2	4606	1754	5244	2667	0.833	11.0	10.9	1101
2.31	99880	38680	16.5	10.3	5075	1939	5812	2949	0.835	10.2	12.3	1441
2.33	110500	42720	16.6	10.3	5537	2129	6387	3243	0.834	9.5	13.8	1854
2.34	116400	44660	16.7	10.4	5777	2222	6681	3384	0.836	9.2	14.6	2073
2.35	122500	46850	16.8	10.4	6031	2325	6999	3544	0.836	8.8	15.5	2343
2.35	125100	47930	16.8	10.4	6141	2375	7140	3621	0.836	8.7	15.9	2478
2.37	141500	53750	17.0	10.5	6803	2646	7975	4038	0.837	8.0	18.2	3292
2.38	146600	55370	17.1	10.5	6998	2721	8222	4154	0.837	7.9	18.9	3545
2.40	159800	60150	17.2	10.6	7516	2940	8887	4492	0.837	7.4	20.9	4336
2.42	180100	67130	17.5	10.7	8283	3257	9877	4983	0.838	6.9	23.9	5655
2.42	183000	67830	17.5	10.7	8383	3291	10000	5034	0.839	6.9	24.3	5809
2.45	203800	75180	17.7	10.8	9151	3619	11010	5543	0.840	6.4	27.5	7401
2.47	226900	82670	18.0	10.9	9962	3951	12080	6058	0.841	6.1	31.1	9240
2.50	249800	90380	18.2	10.9	10750	4289	13130	6583	0.842	5.7	34.8	11350
2.52	274800	98130	18.4	11.0	11580	4629	14240	7108	0.843	5.5	38.8	13720
2.55	299400	106600	18.6	11.1	12400	4987	15340	7669	0.843	5.2	43.0	16460
2.59	341900	120000	19.0	11.3	13740	5555	17160	8553	0.844	4.9	50.2	21390
2.63	392400	135500	19.4	11.4	15260	6202	19260	9560	0.846	4.6	58.9	27870
2.67	450600	153100	19.8	11.6	16960	6922	21610	10680	0.848	4.3	69.4	36130
2.72	518100	173400	20.3	11.7	18850	7736	24250	11960	0.849	4.0	82.0	46760
2.77	597000	196300	20.8	11.9	20970	8640	27230	13370	0.851	3.8	97.1	60230
1.33	12510	410	15.9	2.87	629	57.8	724	90.8	0.858	47.5	0.155	10.7
1.34	15690	538	16.4	3.03	778	75.7	888	118	0.871	38.9	0.207	19.0
1.48	18720	1021	16.5	3.85	930	115	1055	178	0.871	38.3	0.392	23.1
1.49	21600	1203	16.8	3.97	1063	135	1199	209	0.880	33.8	0.466	33.3
1.50	24330	1365	16.9	3.99	1189	153	1346	237	0.880	30.5	0.533	46.1
1.51	27310	1546	17.0	4.04	1323	172	1501	267	0.882	27.6	0.608	62.8
1.52	31520	1794	17.1	4.07	1511	198	1724	309	0.881	24.5	0.714	92.2
1.83	39730	4951	17.7	6.25	1915	381	2128	581	0.888	25.4	1.96	99.4
1.84	46110	5758	17.8	6.28	2198	440	2457	673	0.887	22.4	2.31	149
1.86	54060	6768	17.9	6.33	2541	514	2859	788	0.887	19.7	2.75	227
1.87	61830	7754	18.0	6.39	2869	586	3246	899	0.887	17.7	3.20	322
1.48	21370	645	17.9	3.11	950	84.7	1096	133	0.859	43.9	0.311	21.4
1.49	25500	795	18.3	3.23	1122	104	1287	163	0.868	37.5	0.387	33.8
1.50	28930	913	18.4	3.27	1263	119	1453	187	0.869	33.6	0.448	47.7